

CHARACTERISTICS OF VACCINE HESITANCY AMONG PARENTS IN HILIR PERAK DISTRICT (2016–2024)



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INTRODUCTION

- Vaccine hesitancy, defined as the **delay in acceptance or refusal of vaccination** despite the availability of vaccination services.
- It is a growing public health concern that threatens herd immunity and increases the risk of disease outbreaks.
- This study aimed to describe the characteristics and factors associated with vaccine hesitancy among parents in Hilir Perak District.

MATERIALS AND METHODS

Retrospective Cohort Study

Childhood national immunisation database year 2016- 2024 in Hilir Perak

Sample Size

All vaccine hesitancy records in Hilir Perak were extracted , n = 127

Analysis

Chi Square test / Fisher’s exact test*;
p < 0.05

Outcome Variable

Non-immunised Child:

A child who had not received vaccine included in the Malaysian NIP by 12 months of age

Incomplete Vaccination:

Missing ≥1 NIP vaccine by 12 months of age



RESULTS

127

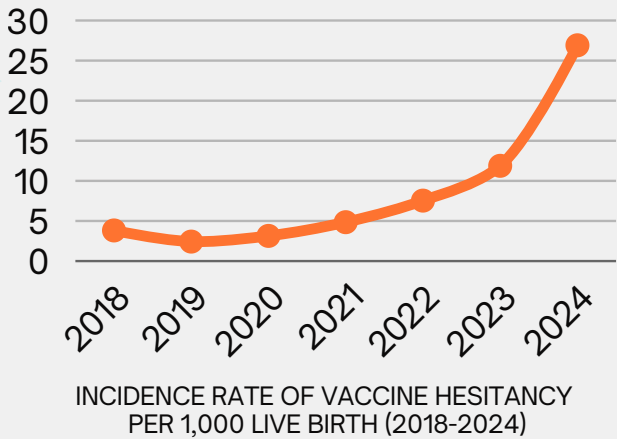
TOTAL VACCINES HESITANCY CASE

69.3%

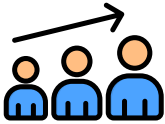
INCOMPLETE IMMUNISATION (n=88)

30.7%

NON-IMMUNISED (n=39)



PARENTS CHARACTERISTICS



MEAN AGE (MOTHER)
33.5 Years old



EDUCATION LEVEL (MOTHER)
61.4% Secondary

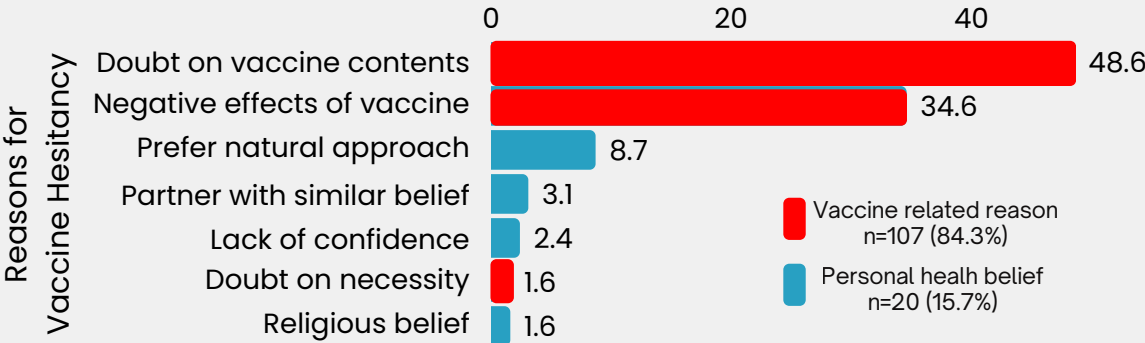


NATIONALITY
100% Malaysian
ETHNICITY
100% Malay
RELIGION
100% Muslim



EMPLOYMENT STATUS (MOTHER)
64.6% Housewife
EMPLOYMENT STATUS (FATHER)
71.7% Non-professional

REASONS FOR VACCINE HESITANCY BY PERCENTAGE (%)



SIGNIFICANT FACTORS ASSOCIATED WITH CHILD IMMUNISATION STATUS

	Factors	Non-Immunised	Incomplete Vaccination	p-value
Maternal Education Level	Tertiary	21 (53.8)	28 (31.8)	0.019
	Secondary	18 (46.2)	60 (68.2)	
Intervention Acceptance	Refused	7 (17.9)	6 (6.8)	0.047*
	Received	32 (82.1)	82 (93.2)	



DISCUSSIONS

Vaccine hesitancy was predominantly selective:

The **predominance of incomplete vaccination over non-immunisation** suggests that many parents accept some vaccines while rejecting others, with safety concerns being the key barrier.

Higher education did not guarantee vaccine acceptance:

Higher maternal education level significantly associated with non-immunised status. Vaccine hesitancy extends beyond educational attainment and requires a deeper understanding of **parental concerns, perceptions and trust.**

Engagement is a critical challenge:

Refusal of intervention was associated with non-immunisation. Targeted parent-centred approaches may be the key to **translating counselling into vaccine acceptance.**

CONCLUSION

The rising trend of vaccine hesitancy in Hilir Perak demands a shift beyond conventional information-based counselling. Targeted, trust-based and effective communication strategies are crucial to improving vaccine acceptance.

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DECLARATION OF CONFLICT

The authors declare no conflicts of interest.

REFERENCES

1.Lim, X.-J., Junus, S., Edwin-Amir, L., & Mohd-Kassim, A. B. (2023). Vaccination refusal in children under 2 years of age in Malaysia 2016–2019. *Asian J Med Health*, 21(6), 1-13.
2.Rumetta, J., Abdul-Hadi, H., & Lee, Y.-K. (2020). A qualitative study on parents’ reasons and recommendations for childhood vaccination refusal in Malaysia. *Journal of Infection and Public Health*, 13(2), 199-203.
3.Othman, N. H., Rajendram, S., & Singh, H. K. B. (2024). Vaccine Hesitancy in Childhood Immunization in the Malaysian Context. *Journal of Pharmacy and Bioallied Sciences*, 16(4), 121-125.